

Physical Principles Behind the Cryogenic Laser Heating Calculator

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1 Introduction

This document summarizes the physical model used in the cryogenic laser heating calculator. The purpose of the application is to estimate the optical and thermal requirements needed to drive a target, located at the end of a fiber-delivered optical path, to a specified temperature rise. The calculation is intended as a transparent engineering estimate. It is designed to identify the relevant scaling with trigger conditions, pulse-train parameters, propagation losses, geometric overlap, absorption, and thermal anchoring. It does not replace an in situ calibration.

The quantity of interest is the temperature rise of the illuminated target. In this setting it is not meaningful to assign a single thermodynamic temperature to the laser field itself at the cold stage. The relevant question is instead how much optical energy is emitted, how much of that energy reaches the end of the fiber, how much is intercepted and absorbed by the target, and how that absorbed power produces a local temperature increase in the presence of finite thermal coupling to a bath. The application is therefore built around the sequence

electrical trigger $\rightarrow$ optical pulse train $\rightarrow$ power delivered to the fiber end
 $\rightarrow$ absorbed power at the target $\rightarrow$ target temperature rise.

The model is intentionally minimal. It retains only the quantities required to estimate the average absorbed power and the resulting thermal response. This is the appropriate level of description for determining what repetition rate, pulse duration, optical loss budget, and thermal parameters are needed to reach a target temperature.

2 Model definition and notation

We denote by f_{rep} the repetition rate of the electrical trigger and by τ_{opt} the optical pulse width. The optical peak power of a single pulse is denoted by P_{peak} . The optical pulse energy is then approximated by

$$E_{\text{pulse}} \approx P_{\text{peak}} \tau_{\text{opt}}. \quad (1)$$

This relation is the starting point of the calculator. Once the laser is triggered reliably and operated within its valid input window, the pulse energy is set by the optical pulse shape and width, while the repetition rate determines how often that energy is delivered.

The corresponding average emitted optical power is

$$P_{\text{avg}} = E_{\text{pulse}} f_{\text{rep}} \approx P_{\text{peak}} \tau_{\text{opt}} f_{\text{rep}}. \quad (2)$$

For the present application, this average power is the central optical quantity because the heating problem is governed by energy deposition over many pulses rather than by the instantaneous peak power alone.

To describe propagation from the source to the end of the fiber, we introduce a total optical loss L expressed in dB. This loss may include distributed attenuation along the fiber and any additional insertion or connector losses. The corresponding transmission factor is

$$\eta_{\text{loss}} = 10^{-L/10}. \quad (3)$$

The average optical power delivered to the fiber end is therefore

$$P_{\text{avg,tip}} = P_{\text{avg}} \eta_{\text{loss}}. \quad (4)$$

Not all of this power is necessarily absorbed by the target. Two additional factors are required. The first is a geometric interception factor η_{geom} , which accounts for beam expansion after the fiber output and the finite size of the illuminated target. The second is an absorptivity factor η_{abs} , which accounts for the fraction of intercepted optical power that is actually absorbed. The average absorbed optical power is then written as

$$P_{\text{abs}} = P_{\text{avg}} \eta_{\text{loss}} \eta_{\text{geom}} \eta_{\text{abs}}. \quad (5)$$

This is the power that enters the thermal model.

The target is described as a lumped thermal mass with temperature $T(t)$, heat capacity C , and thermal conductance G to a bath at temperature T_{bath} . The energy balance is

$$C \frac{dT}{dt} = P_{\text{abs}} - G(T - T_{\text{bath}}). \quad (6)$$

This equation states that the target is heated by absorbed optical power and cooled by thermal relaxation to the bath. It is the simplest self-consistent model that captures both the steady-state temperature rise and the finite timescale required to reach it.

The steady-state limit follows directly from equation (6) by setting $dT/dt = 0$, which gives

$$\Delta T_{\infty} \equiv T_{\infty} - T_{\text{bath}} = \frac{P_{\text{abs}}}{G}. \quad (7)$$

This expression makes the physical dependence clear. The target temperature rises when more optical power is absorbed, and it rises less when the thermal link to the bath is stronger.

The transient solution of equation (6) for constant P_{abs} is

$$T(t) = T_{\text{bath}} + \frac{P_{\text{abs}}}{G} (1 - e^{-t/\tau_{\text{th}}}), \quad \tau_{\text{th}} = \frac{C}{G}. \quad (8)$$

The parameter τ_{th} is the thermal time constant. It determines how quickly the target approaches its asymptotic temperature. In the context of the application, this is the timescale that sets the required burst duration.

If the desired temperature rise is denoted by ΔT_{goal} , then the required illumination time is obtained by solving equation (8) for t :

$$t_{\text{req}} = -\tau_{\text{th}} \ln \left(1 - \frac{\Delta T_{\text{goal}}}{\Delta T_{\infty}} \right), \quad (9)$$

provided that $\Delta T_{\text{goal}} < \Delta T_{\infty}$. If $\Delta T_{\text{goal}} \geq \Delta T_{\infty}$, the requested temperature cannot be reached within this model because the available absorbed power is insufficient relative to the thermal conductance.

Equations (2)–(9) define the core logic of the calculator. The trigger sets the repetition rate, the pulse parameters set the emitted optical energy per pulse, the loss and overlap factors determine how much of that power reaches and is absorbed at the fiber end, and the thermal parameters determine the resulting temperature rise and timescale.

3 Interpretation of the trigger input

For completeness, the application also reports the electrical trigger amplitude in both voltage and dBm when the trigger input is specified in the standard $50\ \Omega$ convention. In that case, if the source power is P_{dBm} , then

$$P = 1\ \text{mW} \times 10^{P_{\text{dBm}}/10}, \quad V_{\text{rms}} = \sqrt{PR}, \quad R = 50\ \Omega. \quad (10)$$

For a sinusoidal source,

$$V_{\text{pp}} = 2\sqrt{2} V_{\text{rms}}. \quad (11)$$

These relations are used only to express the electrical trigger amplitude in convenient units and to compare it against the trigger-input operating window. They are not used to infer a continuous linear scaling of optical power with trigger amplitude. In the present application, once the trigger amplitude is within the valid range, the dominant control parameters for heating are the repetition rate f_{rep} and the optical pulse duration τ_{opt} through equation (2).